

Load Interconnection System Impact Study Report

For Queue Project Q22-2023

West Frankfort 100 kV

May 19, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2023. The Interconnection Customer proposes a 344.1 MW large load (data center) facility interconnecting at the West Frankfort 100 kV substation in the Illinois area. The request was received on March 02, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 644 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,126,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Naval Submarine Base Kings Bay Substation / W Frankfort 100.0 kV

Contingency: Luka / W Frankfort 100.0 kV

Loading: 121.3% (Rating: 166 MVA)

DFAX: 100.0%

Pre-project loading: 85.8%

2. Facility: W Frankfort / Luka Substation 100.0 kV

Contingency: Naval Submarine Base Kings Bay / West Frankfort 100.0 kV

Loading: 117.8% (Rating: 171 MVA)

DFAX: 100.0%

Pre-project loading: 83.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Naval Submarine Base Kings Bay / W Frankfort 100.0 kV	Rebuild line with higher-capacity conductor	\$2,223,000	64% (\$1,416,000)
West Frankfort / Luka 100.0 kV L	Rebuild line with higher-capacity conductor	\$1,116,000	64% (\$710,000)
	Total allocated to project		\$2,126,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.